

MATH 3210-004 Final Guide

Instructor: Alp Uzman

Subject to Change; Last Updated: 2026-04-20 22:22:42-06:00

1 Logistics

Our final exam is scheduled for Friday, April 24, at the usual class location (**WBB** 617) from 8:00 AM to 10:00 AM. *Note that the exam starts earlier than the regular meeting time.*

Arrival Time Please plan to arrive and be settled in the classroom by 7:55 AM on the day of the exam.

Exam Materials At the start of the exam you will receive two separate documents—a policy document and an exam booklet containing all the test questions. Both documents will be collected by the end of the exam. Both documents will be collected by the end of the exam. Samples of both of these documents are available at

[https://github.com/AlpUzman/
MATH_3210_001_SPRING_2025/
tree/main/Exams](https://github.com/AlpUzman/MATH_3210_001_SPRING_2025/tree/main/Exams)

You may use the sample exam booklet to practice for the exam.

Office Hours My office hours are scheduled by appointment. If you'd like to set up a meeting, please send me a request specifying your available times. Additionally, provide sufficient notice so I can accommodate your request in my schedule.

2 Exam Scope

Final exam will be cumulative, although the emphasis will be on the following topics:

- the fundamental theorems of calculus,
- Taylor approximation, including various Mean Value Theorems from integral and differential calculus.

The exam may include problems—verbatim or modified—from previous problem sets, exercises mentioned in class, follow-up exercises from the selected solutions document on Canvas, or the textbook.

What you can assume in the exam During the exam you may assume all statements proved in class. Additionally, you may assume statements proved in the textbook, except when they appear as a problem in a previous problem set, or as an exercise in class. Exceptions to these general rules are:

- You may assume (the corrected statements from) PSet 1.
- You may assume the statements in problem 2 of PSET3; namely you may assume the limit laws for sequences in the exam.
- You may assume the exercise regarding properties of limit inferior and limit superior mentioned in class on 1/30/2026.

- You may assume that for real valued functions defined on a compact interval, continuity is equivalent to uniform continuity.
- You may assume (the corrected statements from) problem 5 from PSet 4; this problem involves different characterizations of continuity.
- You may assume (the corrected statements from) problems 2 and 3 from PSet 5.
- You may assume (the corrected statements from) problems 1 to 4 from PSet 6.
- You may assume (the corrected statements from) problems 5 and 6 from PSet 7.
- You may assume (the corrected statements from) problem 2 from PSet 9.
- You may assume (the corrected statements from) problem 1 from PSet 10.